CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY

College of Computer Studies

Bachelor of Science in Information Technology

Software Project Management Plan (SPMP v2.0)

PlayIT

An Offline-First Gamified Early Literacy Mobile Application

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Document Revision History

Version	Date	Primary Author(s)	Management Plan Refactoring Description
1.0	May 4, 2026	Project Manager	Initial SPMP baseline developed for IT332 Capstone 1.
2.0	September 26, 2026	Project Manager & Capstone Team	**Comprehensive SPMP Refactoring Based on Weeks 1–2 MVP Validation Findings:** • **Project Scope**: Formally added the Tutoring & Speech Judgment Layer, Three-Stage Audio Production Pipeline, Learner Autonomy scaffolding, and Local Telemetry Logging/Export; prioritized Chapter 1 (*m, s, a, i*) as the Round 2 vertical slice. • **Activities & Schedule**: Re-baselined the Weeks 3–9 implementation and testing timeline to incorporate phonetic audio re-mastering, Vosk grammar spikes, and Round 2 Silent-Observer validation. • **Work Breakdown Structure (WBS)**: Expanded into 9 structured work packages covering pedagogy design, audio pipeline, ASR spike/judge, tutoring layer, independence features, telemetry, testing, Round 2 validation, and documentation. • **Resource Allocation**: Allocated compute and tooling for local Kokoro-82M neural TTS, Chatterbox-Turbo voice cloning, and physical Android test tablets. • **Roles & Responsibilities**: Re-assigned 6 specialized functional roles across the 5 team members. • **Risk Management**: Expanded the Risk Register to 8 targeted engineering and pedagogical risks (R1–R8) with concrete mitigation and contingency protocols.
2.1	October 5, 2026	Capstone Team (Claude review)	Corrections for adviser review: status changed to draft; findings table separates implemented from planned work; teacher names replaced by codes; Room schema v4; Milestone 1 marked submitted and revised.
2.2	October 9, 2026	Capstone Team (Claude, sprint synchronization)	Records the Oct 9–10 refactoring sprint on branch `refactor/hear-say-it` (cards 18–24, 03b, 15; 364 unit tests passing): §1.2 findings rows updated with what shipped; §3 adds the sprint as a completed internal milestone and updates Milestones 2–4 honestly (shipped vs still planned). Cross-references `MVP_Validation_Findings_and_Refactoring_Priorities_Filled.md` and `engineering-package/13_MASTER_TASKS.md`.
2.3	October 10, 2026	Capstone Team (Claude, submission review)	Review for the IT411 submission: §1.2 design-impact cells name the shipped classes (`AudioResolver`, `SpeechValidator`) and the planned ones; word-bank teacher sign-off marked pending; §1.2 sources match the validation data; §3 records the automated batch validation of the sprint (364 tests, 41 screenshots, debug APK) and the open parts of Milestone 3; §7.1 describes the asset release gate as it works today; WBS tree without stray Markdown.
2.4	October 11, 2026	Capstone Team	Issued for adviser review and evaluation: the front matter states the submission status and the members' full names; Milestone 1 records the resubmission. No schedule change.

Version	Date	Primary Author(s)	Management Plan Refactoring Description
2.5	October 14, 2026	Capstone Team (Claude, card 30)	Synchronized with debug APK G (`89604d7`) for the October 14 submission: §1.2 records the ElevenLabs clips (AM, TUB, YAM, short /i/), the mouth cue in Say It only and the shipped mouth pictures; §3 adds the Oct 10–13 phone-test sprint (cards 28 to 29b, 409 unit tests) and updates Milestones 1 and 2; R1, R5 and R7 cover the ElevenLabs fallback, its Free-plan license and the short /a/ still to make; §4.1 lists ElevenLabs as a build-time tool. No change to the later milestones.

1. Project Overview & Scope Refactoring

1.1 Project Purpose & Executive Summary

PlayIT is an offline-first native Android educational mobile application designed to instruct Grade 1 Filipino learners in English phonemic awareness, speech production, and CVC word decoding following the Department of Education's (DepEd) Marungko Approach. Following the formative MVP Field Validation conducted in Weeks 1–2 (evaluating N = 25 participants: 16 early learners, 5 parents or supervising teachers, and 4 certified DepEd teachers), this SPMP v2.0 recalibrates project execution, resource allocation, and team milestones to directly address the empirical findings and advisory directives prior to full system rollout.

1.2 Interconnected Deliverables Traceability Baseline

In accordance with IT411 academic guidelines, the Weeks 1–6 project lifecycle follows an unbroken evidence-to-implementation chain:

MVP Validation → Findings & Feedback → Refactored Requirements (SRS v3.0) → Updated System Design (SDD v2.0) → Updated Project Plan (SPMP v2.0)

The specific empirical findings and their resulting impact on project scope and management are summarized below:

Empirical Validation Finding (Weeks 1–2)	Stakeholder Feedback Source	SRS v3.0 Requirement Impact	SDD v2.0 Architectural Impact	SPMP v2.0 Scope & Management Impact
Vocal schwa / letter-name intrusion in synthesized phoneme models (e.g., "ma" instead of pure /m/ [m:]).	`PED-08` (50% flagged by DepEd teachers); Teachers T-1 and T-2 comments.	**FR-02, NFR-AUD-01:** Mandated pure acoustic phoneme models and runtime sequence assembly.	Implemented: pure phonemes (`ph_m.wav`, `ph_s.wav`, `ph_i.wav`) kept separate from key-word clips (`kw_mouse.wav`) through `AudioResolver`; `AudioPlayer` uses `SoundPool` for short clips. Planned: `AudioPlaybackManager`, `AudioComposer`.	**WBS 2 (Audio Pipeline):** Established 3-stage release gates (Gates 1–3) using Kokoro-82M and Chatterbox-Turbo, with ElevenLabs Text to Speech for the clips they could not make clean (AM, TUB, YAM, short /i/).
Child speech hesitation at mic; lack of active listening indicator.	Facilitator notes; Perceived Ease lowest affective rating (7 of 16 children did not choose the top face).	**FR-03:** Mandated 4 mic states, live RMS ripple, tap-to-listen ≤ 100 ms, no heart loss.	Implemented 2026-10-09: 4 mic states (`MicStatus`, `MicButton`, time-based ripple). Planned: `AudioRecord` PCM loop for the voice-driven ripple.	**WBS 4 & 5:** Added dynamic visualizer and mascot prompting tasks to sprint schedule.

Empirical Validation Finding (Weeks 1–2)	Stakeholder Feedback Source	SRS v3.0 Requirement Impact	SDD v2.0 Architectural Impact	SPMP v2.0 Scope & Management Impact
Untested speech recognition agreement on child vocal productions.	Empirical validation gap; General Objective 2 requirement.	**NFR-ASR-01:** Established ≥ 80% agreement target and ≤ 15% false reject ceiling.	Implemented: `SpeechValidator` (word-mode judge with per-letter foil grammars and error types) in the role of the planned `SayItJudge`.	**WBS 3 (ASR Spike):** Scheduled Chapter 1 acoustic calibration spike and threshold tuning.
5 invalid CVC words in Blend It containing unintroduced vowel teams.	Post-pilot pedagogical review; decodability violation.	**FR-13:** Replaced AIM, BEE, TOY, BOY, ZOO with AM, SUM, TUB, YAM, ZIP.	Implemented 2026-10-09: `BLEND_IT_WORD_SEEDS` reseeded on every open (card 15); the AM, TUB and YAM clips shipped 2026-10-12 (card 15b); teacher confirmation and the pictures for AM, TUB, YAM and ZIP pending.	**WBS 1 (Pedagogy Design):** Curated the 33-word decodable bank; teacher sign-off pending (Instrument A, Part 2).
Desire for classroom reading stations on shared Android tablets.	`PED-12` (4 of 4 teachers: ready as a supplementary classroom tool); shared classroom devices need separate profiles.	**FR-14:** Formalized multi-profile management (up to 6 child profiles per device).	`ProfileEntity` and `SessionManager` exist (multi-profile implemented).	**WBS 4:** Prioritized multi-profile state scoping and avatar switcher in sprint deliverables.
Missing telemetry and latency metrics during field validation.	Validation gap (no Round 1 latency or agreement data); teachers asked for progress tracking.	**FR-NEW-TEL:** Mandated offline event logging and PIN-gated CSV/PDF export.	Planned: `TelemetryLogger` and `CsvExportManager`.	**WBS 6:** Dedicated work package for offline telemetry recording and local export.
Hearing and motor accessibility gaps (hearing score 60%, corner toggles).	`ACC-06` (3 of 5 Yes); `ACC-07` (4 of 5 Yes).	**NFR-ACC-01, NFR-ACC-02:** Sound captions, visual lip cues, ≥ 64 dp touch targets.	Implemented: `ArticulationCue` with 9 mouth pictures (card 24b), shown in Say It from the second miss and not in Hear It (card 28c); captions built but not shown in the child view (card 28); adaptive `PlayItDimens` tokens with 64 dp floors and `LessonScaffold`.	**WBS 5:** Sourced visual mouth articulation assets and updated UI layout modifiers.

1.3 Scope Baseline & Phased Implementation Strategy

- **Core Curricular Track:** 26 letters of the English alphabet structured across 7 chapters.
- **Round 2 Vertical Slice Strategy:** To de-risk speech recognition and audio modeling before scaling to all 26 letters, **Chapter 1 (*m, s, a, i*) serves as the primary vertical slice** for full tutoring integration, audio gate clearance, and Round 2 field testing.
- **Deferred Scope (P3):** Digraphs (e.g., *sh, ch, th*) and consonant clusters are documented as data-driven architectural extensions scheduled for post-capstone releases.

2. Work Breakdown Structure (WBS) & Deliverables

The refactored project scope is organized into 9 distinct Work Packages (WBS 1 to 9):

```
1.0 PlayIT System Refactoring & Implementation
  |-- 1.1 WBS 1: Pedagogy & Curriculum Design
  |     |-- 1.1.1 Curate 26-Letter Lesson Scripts (Chapter 1 Priority)
  |     |-- 1.1.2 Finalize 33 Decodable CVC Word Bank (Replace 5 Invalid Words)
  |     +-- 1.1.3 Construct Spoken Carrier & Formative Feedback Phrase Library
  |-- 1.2 WBS 2: Neural Audio Pipeline & Gate Auditing
  |     |-- 1.2.1 Deploy Local Kokoro-82M & Chatterbox-Turbo Tooling
  |     |-- 1.2.2 Execute Gate 1 Internal Acoustic Audit (Purity, Duration, SNR)
  |     |-- 1.2.3 Execute Gate 2 Vosk ASR Technical Acceptance
  |     +-- 1.2.4 Compile Formal JSON Asset Manifest (manifest.json)
  |-- 1.3 WBS 3: ASR Calibration Spike & Speech Judge
  |     |-- 1.3.1 Conduct Chapter 1 Vosk Grammar Spike (<target>, foils, [unk])
  |     |-- 1.3.2 Implement SayItJudge Decision & Error-Tagging Engine
  |     +-- 1.3.3 Calibrate Confidence Thresholds Against Child Pilot Audio
  |-- 1.4 WBS 4: Tutoring Layer & State Machine
  |     |-- 1.4.1 Implement LessonEngine & TutorPolicy FSM
  |     |-- 1.4.2 Implement AudioComposer Dynamic Sequence Builder
  |     +-- 1.4.3 Integrate Room Schema v4 (Profile, LetterProgress, Telemetry)
  |-- 1.5 WBS 5: Learner Autonomy & Pediatric Accessibility
  |     |-- 1.5.1 Build Mascot Dual-Coding Demos & 10s Idle Re-Prompter
  |     |-- 1.5.2 Build MicStateVisualizer with Live RMS Audio Ripple
  |     +-- 1.5.3 Integrate ArticulationCue Guides & Global 64dp Touch Modifiers
  |-- 1.6 WBS 6: Offline Telemetry & Local Export Engine
  |     |-- 1.6.1 Implement Real-Time Event Logger (SystemClock.elapsedRealtime)
  |     +-- 1.6.2 Implement PIN-Protected Local CSV & PDF Exporter
  |-- 1.7 WBS 7: Verification & Software Test Documents (STD)
  |     |-- 1.7.1 Formulate Test Plan & Unit Test Suites (TC-AUD, TC-MIC, TC-ASR)
  |     +-- 1.7.2 Execute Internal Dry Run with 3–5 Early Learners
  |-- 1.8 WBS 8: Round 2 Field Validation Execution
  |     |-- 1.8.1 Administer Silent-Observer Protocol with Target Cohort (N=30)
  |     +-- 1.8.2 Conduct Gate 3 Pedagogical Teacher Audit on Audio Assets
  |-- 1.9 WBS 9: Capstone Documentation & Dissemination
        |-- 1.9.1 Compile Refactored SRS v3.0, SDD v2.0, and SPMP v2.0
        +-- 1.9.2 Draft Final Midterm Validation & Implementation Report
```

3. Activities, Milestones & Revised Master Schedule

The revised master schedule spans Weeks 3 through 9 of the First Semester, Academic Year 2026–2027:

Milestone / Deliverable	Week	Target Dates (2026)	Primary Activities & Work Packages Involved	Status / Verification Gate
Milestone 1: Refactored Specifications Submission	**Week 3**	Sept 21 – Sept 26	Finalize and submit refactored **SRS v3.0**, **SDD v2.0**, **RTM v3.0**, and **SPMP v2.0** incorporating MVP validation findings.	**Submitted; revised 2026-10-05, 2026-10-11 and 2026-10-14 for adviser review and evaluation**
Milestone 2: ASR Spike & Chapter 1 Audio Gate Clearance	**Week 4**	Sept 28 – Oct 3	Conduct Vosk grammar spike on *m, s, a, i*; synthesize Chapter 1 phonemes via Kokoro/Chatterbox; execute Gates 1 & 2; verify decodable word bank.	**In Progress** (Spike Decision Gate); held /m/ and /s/ and the short /i/ released, short /a/ pending; word list implemented, teacher audit pending

Milestone / Deliverable	Week	Target Dates (2026)	Primary Activities & Work Packages Involved	Status / Verification Gate
Milestone 3: Tutoring Engine & AudioComposer Integration	**Week 5**	Oct 5 – Oct 10	Implement `LessonEngine`, `TutorPolicy` FSM, `AudioComposer`, and Room Schema v4 migration; synthesize remaining phoneme assets (Ch 2–7).	**In progress:** `TutorPolicy` shipped as a stateless policy; `LessonEngine`, `AudioComposer`, Room schema v4 and the Ch 2–7 phonemes are still planned (SDD §2.0)
Internal sprint: UI fit, mic states, corrections, accessibility	**Week 5**	Oct 9 – Oct 10	Cards 18–24, 03b and 15 on `refactor/hear-say-it`: adaptive layout on 4 device sizes, 4-state mic, spoken Say It corrections, decodable word list, responsive map, purposeful effects, captions and mouth cues (see `engineering-package/13_MASTER_TASKS.md`).	**Shipped and validated** (automated batch validation `c07206e`: 364 unit tests passing, 41 Roborazzi screenshots on 4 sizes, debug APK `playit-debug-B-eaf8634.apk`)
Internal sprint 2: phone-test fixes and asset releases	**Week 6**	Oct 10 – Oct 13	Cards 28, 28b, 28c, 28d, 15b, 09c, 29 and 29b on `refactor/hear-say-it`, from the team's phone tests on a Samsung Galaxy A21s: Hear It ends on the sound and shows only the letter, the mouth cue moves to Say It, a lesson screen shows the right picture after "next letter", ElevenLabs clips for AM, TUB, YAM and the short /i/ with the Parent Zone credit, and the redrawn mascot and avatars.	**Shipped:** 409 unit tests passing at `89604d7`; debug APK G (`playit-debug-G-89604d7.apk`) is the submission build. Every refactor card on the task list is done except card 16, which card 20 replaced.
Milestone 4: Autonomy Features, Telemetry & Gate 3 Teacher Audit	**Week 6**	Oct 12 – Oct 17	Implement `MicStateVisualizer`, 10s idle re-prompter, 64dp touch tokens, and CSV exporter; conduct Gate 3 teacher audit on audio; execute internal dry run (N = 5).	**Partly done:** mic states, idle re-prompt and 64 dp tokens shipped (sprint, Oct 9–10); CSV exporter and Gate 3 audit pending
Milestone 5: Round 2 Field Validation Execution	**Week 7**	Oct 19 – Oct 24	Deploy `playit-v2-debug.apk` to physical test tablets; conduct multi-stakeholder testing (N ≥ 30) under the Silent-Observer protocol.	Field Empirical Data Collection
Milestone 6: Empirical Analysis & Feature Freeze	**Week 8**	Oct 26 – Oct 31	Analyze Round 2 telemetry (ASR agreement, latency, recall retention, SUS); declare feature freeze; prepare Software Test Documents (STD).	Milestone Review Gate
Milestone 7: Final Midterm Submission & Oral Defense	**Week 9**	Nov 2 – Nov 7	Finalize Capstone 2 Midterm Report, verified STD execution logs, and live system demonstration.	Midterm Defense

4. Resource Allocation & Infrastructure

4.1 Development & Processing Infrastructure

- **Workstations:** High-performance local development machines running Windows 11 with WSL2 (Ubuntu 22.04 LTS) for neural audio synthesis pipelines.

- **Neural TTS Engine:** Kokoro-82M PyTorch pipeline (Apache-2.0 license) executed in local Python environments; Chatterbox-Turbo (MIT license) for held continuous consonant synthesis; ElevenLabs Text to Speech (cloud, Free plan, used at build time only, so the app stays offline) for the clips the local tools could not make clean.
- **Acoustic Analysis:** Praat phonetics toolkit for automated formant, duration, and schwa verification.

4.2 Target Hardware & Test Fleet

- **Physical Test Devices:** Fleet of 4 dedicated Android tablets (Samsung Galaxy Tab A7/A8, Android 11/12) and 2 smartphones (Android 10/13) representative of low-cost hardware used in Philippine public elementary schools.
- **Audio Accessories:** Calibrated external sound level meter to monitor ambient noise (≤ 40 dB) during testing.

4.3 Software Libraries & Frameworks

- **Programming Language:** Kotlin 1.9.22 (100% pure Kotlin in domain layer).
- **UI Framework:** Jetpack Compose 1.5.4 with Material 3 design tokens.
- **Offline ASR Engine:** Vosk Android SDK v0.3.47 bundled with `vosk-model-small-en-us-0.15` (approx. 40MB packaged in `assets/model/`).
- **Database & Persistence:** Room 2.6.1 + SQLite.
- **Document Generation:** Android native `PdfDocument` API.

5. Team Roles and Responsibilities

To ensure clear accountability, 6 key project functions are allocated across the 5 capstone team members:

Functional Role	Designated Lead	Primary Responsibilities & Accountabilities	Mapped Work Packages
Project Manager & Scrum Master	Team Lead / Member 1	Master schedule maintenance, sprint tracking, adviser liaison, risk register oversight, transmittal letter coordination.	WBS 8, WBS 9
Pedagogy & Curriculum Lead	Member 2	DepEd Marungko sequence alignment, 26-letter lesson scripts, decodable CVC word bank curation, teacher interview liaison.	WBS 1, WBS 8
Neural Audio & Acoustic Lead	Member 3	Kokoro-82M / Chatterbox synthesis pipelines, Gate 1 acoustic audits, asset manifest compilation, SoundPool integration.	WBS 2
Speech Recognition (ASR) Lead	Member 4	Vosk SDK integration, per-letter dynamic grammar compilation, ASR agreement calibration, latency optimization.	WBS 3
Android & Architecture Lead	Member 5	Clean Architecture maintenance, Compose UI, `LessonEngine`, `TutorPolicy` FSM, Room Schema v4 migration, telemetry logging.	WBS 4, WBS 5, WBS 6
QA & Validation Lead	Member 1 (Dual Role)	Software Test Documents (STD), automated regression suites, Gate 2 technical verification, Round 2 Silent-Observer field logistics.	WBS 7, WBS 8

6. Comprehensive Risk Management & Mitigation Matrix

The following risk management matrix addresses 8 critical technical, pedagogical, and operational risks identified following MVP validation:

Risk ID	Risk Description	Category	Likelihood	Impact	Proactive Mitigation Strategy	Reactive Contingency Plan	Owner
R1	Neural TTS cannot synthesize pure stop sounds without schwa ("buh" for /b/).	Technical / Audio	**High**	**High**	Continuous sounds generated via Kokoro/Chatterbox; short stops sourced directly from human reading specialist recordings; mandatory Gate 1 acoustic screening. Where Kokoro could not make a clean clip (the word TUB, the short /i/), ElevenLabs Text to Speech was used (team decisions 2026-10-11 and 2026-10-12).	If an AI clip fails Gate 3 twice, replace immediately with a validated human studio recording.	Audio Lead
R2	Vosk cannot reliably discriminate target phonemes from foils in isolated sounds.	Technical / ASR	**Medium**	**High**	Conduct Chapter 1 Vosk grammar spike; compile per-letter grammars (`<target>`, foils, `[unk]`); tune confidence thresholds on pilot child audio.	Evaluate the key word for vocal scoring (e.g., *"mouse"*) while keeping the pure sound model for instruction; consult adviser.	Speech Lead
R3	Grade 1 children cannot navigate English-only instructions independently.	Pedagogical / UX	**Medium**	**High**	Implement dual-coding: short spoken carrier prompts (≤ 8 words) paired with vector icons; mascot first-use demos; 10s idle re-prompt.	Deploy an optional auxiliary audio track with Filipino/Cebuano carrier explanations for remedial centers.	Pedagogy Lead
R4	ASR false rejections induce frustration in young children without an adult present.	User Affect / Gamification	**Medium**	**High**	Completely eliminate heart penalties in Say It; implement progressive 3-tier prompt ladder; enforce ≤ 15% false reject ceiling.	Broaden acoustic grammar tolerances or widen phonetic acceptance windows for Philippine English variants.	Android Lead

Risk ID	Risk Description	Category	Likelihood	Impact	Proactive Mitigation Strategy	Reactive Contingency Plan	Owner
R5	Producing scripts, art, and audio for all 26 letters causes schedule overruns.	Project Schedule	**High**	**Medium**	Implement Chapter 1 (*m, s, a, i*) as the complete vertical slice for Round 2; establish standardized JSON script templates. Status 2026-10-13: /m/, /s/ and the short /i/ are released; the short /a/ still needs the team member's recordings or a paid ElevenLabs plan (the account's Free Tier is no longer available) before Round 2.	Move Chapters 2 to 7 production to sprint immediately following Round 2 validation.	Project Manager
R6	Administrative delays in parental consent and school scheduling for Round 2.	Operational / Logistics	**Medium**	**High**	Distribute parental consent forms and school coordination letters during Week 4; pre-book testing sessions early.	Shift Round 2 by 1 week, utilize community reading centers, and compress data analysis window.	QA / Validation Lead
R7	Third-party neural voice models present open-source licensing ambiguities.	Legal / Compliance	**Low**	**Medium**	Strictly enforce Apache-2.0 (Kokoro-82M) and MIT (Chatterbox-Turbo) licenses; log licenses in asset manifest; avoid GPL runtime dependencies. The ElevenLabs clips are Free-plan audio: non-commercial only, credited as "elevenlabs.io" in the Parent Zone (`VoiceCredit.kt`).	Revert exclusively to human voice recordings if licensing disputes arise. A commercial release re-makes the ElevenLabs clips on a paid plan or replaces them, because Free-plan audio cannot become commercial later.	Audio Lead
R8	Team availability drops during midterm examination period.	Team Resources	**Medium**	**Medium**	Cross-train pairs on critical paths (Audio+Speech; Android+QA); maintain buffer days prior to milestones.	Reassign non-critical tasks during weekly sprint check-ins to protect primary deliverables.	Project Manager

7. Configuration Management & Quality Assurance

7.1 Configuration Management & Asset Versioning

- **Source Code Repository:** Git version control hosted on GitHub (`refactor/hear-say-it` feature branch merged via reviewed pull requests).
- **Release Tagging:** Every deployable APK build tested in the field is assigned a semantic tag (e.g., `v2.0.0-rc1-round2`).
- **Asset Manifest Control:** Audio and image assets enter `app/src/main/assets/` only through a dated release manifest (`docs/audio-release/<date>/manifest.json`, `docs/image-release/<date>/manifest.json`) that records each file's SHA-256 hash after the team approves it on a review page. Unit tests check the shipped pictures against the image manifest (`PictureAssetsTest`), and the Gate 3 teacher audit of every shipped clip is required before merging to `main`.
- **Document Synchronization:** SRS v3.0, SDD v2.0, RTM v3.0, and SPMP v2.0 are versioned and released concurrently as an integrated specification bundle.

7.2 Quality Assurance & Verification Protocol

1. **Continuous Integration (CI):** Every commit to the repository triggers automated Gradle verification (`./gradlew testDebugUnitTest`) executing unit tests across DAOs, ViewModels, and state machines.
2. **Audio Verification Gates (Gates 1, 2, 3):**
   - *Gate 1 (Internal):* Waveform duration and acoustic purity audit via Praat.
   - *Gate 2 (Technical):* Automated Vosk batch recognition testing.
   - *Gate 3 (Pedagogical):* Blind listening audit by ≥ 3 DepEd reading teachers.
3. **Round 2 Field Validation (Silent-Observer Protocol):** Field testing conducted with N ≥ 30 Grade 1 learners to verify unprompted completion (≥ 85%) and true phonetic recall (≥ 70%).